

11. **(a)** Explain the significance of quantum numbers.
(b) Draw the atomic structure of Silicon (Si) and Germanium (Ge). **(5 + 5)** **(CO1)**
12. **(a)** Define ionic bond and covalent bond. Explain the formation of NaCl using the Lewis dot method.
(b) Define buffer solution. Give two examples and mention their applications. **(5 + 5)** **(CO1)**
13. **(a)** Explain the construction and working of a lead-acid battery.
(b) State the differences between electrolytic cells and galvanic cells. **(5 + 5)** **(CO2)**
14. **(a)** Explain the mechanism of rusting of iron.
(b) Write a short note on the sacrificial anode method of corrosion prevention. **(5 + 5)** **(CO2)**
15. **(a)** Define hard water. Explain the ion-exchange process for softening hard water with a neat diagram.
(b) Calculate the hardness of water if 100 ml of water contains 0.01 g of CaCO_3 . **(6 + 4)** **(CO3)**
16. **(a)** Define elastomer. Explain the preparation and applications of Neoprene.
(b) What are liquid crystals? Give two examples and mention their applications. **(5 + 5)** **(CO4)**
17. **(a)** Define deforestation. Explain its causes and effects.
(b) Write a short note on the greenhouse effect and its impact on the environment. **(5 + 5)** **(CO5)**
18. **(a)** Define green chemistry. Mention its principles and benefits.
(b) Explain the causes and effects of water pollution. **(5 + 5)** **(CO5)**